Kala Chakra (*kāla cakra*)

The Wheel of Time — Odia: written *kāla cakra*

Automating the KP Horary Workflow

From raw astrological export to star/sub lords, cusp numbers, and the step-wise power calculation

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Method:	Krishnamurti Paddhati (KP), East Indian style
Deliverables:	Python package, JavaScript app, Excel pipeline
Date:	July 6, 2026

This document describes the computational method exactly as implemented. Every rule was reconstructed from the author's manual workflow (the `Guide_Step1`–`Guide_Step6` sheets and hand-drawn notes) and validated against a corpus of previously hand-calculated charts. It is intended as a durable reference for future maintenance and debugging.

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1 Introduction

Horary astrology answers a specific question — "Will I get this job?", "Will the admission come through?" — from a chart cast for the moment the question is posed, at a given place and time. Under Krishnamurti Paddhati (KP), the answer is read not from the signs alone but from a nested lordship scheme: each point in the zodiac has a *star lord* (the ruler of its nakshatra) and a *sub lord* (a finer subdivision), and the relationships between the planets, their lords, and the twelve house cusps are combined into a numeric "power" expression that the astrologer interprets.

Performed by hand, one chart takes 15–25 minutes of careful table lookups and arithmetic, and the work is error-prone: a single mis-read table row or an off-by-one house boundary silently corrupts everything downstream. **Kala Chakra** automates the entire sequence, reducing a chart to 2–3 minutes with no arithmetic slips, while showing every intermediate step so the astrologer retains full interpretive control.

Scope. The system automates everything up to and including the power calculation. It does *not* interpret the result or cast dasha periods — those remain the astrologer's work. Planetary and cuspal longitudes are taken as given from professional astrology software; Kala Chakra does not compute ephemerides.

2 The Six-Step Manual Workflow

The automation mirrors the author's own six-step process, so that each block can be checked against the manual method it replaces.

1. **Capture the raw chart.** The astrology software (Solar Fire) produces a Chart Analysis Report: house cusps and planetary longitudes in degrees, minutes and seconds, plus a large amount of unrelated material.
2. **Clean and lay out.** Keep only the twelve cusps and nine KP points. The node `Nod` is renamed `Ra` (Rahu); `Ke` (Ketu) is added at the same degree in the opposite sign. Retrograde markers are dropped. The retained points are arranged in the fixed East Indian chart layout.
3. **Record the query context.** Date, time and location are noted at the centre of the chart.
4. **Assign star and sub lords, placements and ownership.** For each cusp and planet, look up (here: *compute*) the star lord and sub lord. Assign each planet the cusp it occupies, and the cusp numbers it owns.
5. **Write the cusp-number vectors.** For each planet, an ordered vector [placement, ownership…]; for the nodes, a special dispositor-based vector.
6. **Power calculation.** Combine placements and lordships into a numerator/denominator "power" expression, merge like terms, and render the final ascending line for each planet.

3 The Computational Core

3.1 Exact arithmetic in arcseconds

Every longitude is stored as an integer number of arcseconds from 0° Aries:

$$\lambda \;=\; 3600\,d \;+\; 60\,m \;+\; s \;+\; 108{,}000 \times (\text{sign index}), \qquad 0 \le \lambda < 1{,}296{,}000.$$

Because all KP boundaries fall on exact arcsecond values, integer arithmetic makes the computation exact — there is no floating-point rounding anywhere in the chain. A full circle is 1,296,000″; a sign is 108,000″; a nakshatra is 48,000″ (13°20′).

3.2 Star lord and sub lord from Vimshottari proportions

The 27 nakshatras cycle through nine lords in the fixed Vimshottari order Ketu, Venus, Sun, Moon, Mars, Rahu, Jupiter, Saturn, Mercury. The **star lord** of a longitude is simply the lord of the nakshatra it falls in:

$$\text{star lord} \;=\; \text{VIM}\big[\,\lfloor \lambda/48{,}000 \rfloor \bmod 9\,\big].$$

The **sub lord** subdivides each nakshatra in proportion to the Vimshottari dasha years (Ke 7, Ve 20, Su 6, Mo 10, Ma 7, Ra 18, Ju 16, Sa 19, Me 17; total 120). The width of a sub within a nakshatra is

$$\text{width}(\ell) \;=\; \frac{\text{years}(\ell)}{120} \times 48{,}000 \;=\; 400 \times \text{years}(\ell) \text{ arcseconds},$$

and the subs run in Vimshottari order *starting from the star lord*. Walking these cumulative widths from the start of the nakshatra locates the sub. Boundaries are half-open, $[\text{from}, \text{to})$, matching the printed KP tables.

Key design decision. The 249-subdivision KP table is *generated*, not stored. It follows deterministically from nine integers (the dasha years), so there is no lookup file to transcribe or keep in sync. The generated table was checked row-by-row against the scanned reference tables, including the author's father's handwritten Taurus corrections, with zero discrepancies.

3.3 Placement and ownership

Placement. House h spans the arc from cusp h to cusp $h{+}1$ (in zodiacal order). A planet at longitude λ occupies house h when $(\lambda - c_h) \bmod 1{,}296{,}000 < (c_{h+1} - c_h) \bmod 1{,}296{,}000$. This modular test handles the wrap-around at 0° Aries and the case of two cusps in one sign automatically.

Ownership. Each planet rules one or two signs (Table 1); it owns the cusp numbers that fall in those signs. When one sign contains two cusps, its lord owns both. When a sign contains *no* cusp (because a neighbouring sign holds two), the whole sign lies inside a single house span, and its lord owns that house — the **span rule**. Charts triggering the span rule are flagged with an informational note.

Nodes. Rahu and Ketu own no sign. They act through their *dispositor* (the lord of the sign they occupy). The cusp-number vector for a node is

$$[\,\text{dispositor's placement},\ \text{node's placement},\ \text{dispositor's ownerships}\ldots\,].$$

Table 1: Sign lordship used throughout (traditional KP rulership).

Planet	Signs owned
Sun (Su)	Leo
Moon (Mo)	Cancer
Mars (Ma)	Aries, Scorpio
Mercury (Me)	Gemini, Virgo
Jupiter (Ju)	Sagittarius, Pisces
Venus (Ve)	Taurus, Libra
Saturn (Sa)	Capricorn, Aquarius
Rahu / Ketu	None — act through the dispositor

4 The Power Calculation

The power calculation is the crux of the method. For each planet we identify its Star lord and Sub lord (of the *planet*, not the cusp), and build a fraction whose terms are cusp numbers raised to fixed powers. The general form, transcribed from the author's notation, is

$$(\text{Planet},\ \text{Star},\ \text{Sub}) \;=\; \frac{(\text{Planet})^{2}(\text{own})^{1} \;-\; (\text{Star})^{4}(\text{own})^{3} \;-\; (\text{Sub})^{6}(\text{own})^{5}}{(\text{Star})^{7}(\text{Sub})^{10} \;-\; (\text{Star})^{8}(\text{Sub})^{11} \;-\; (\text{Star})^{9}(\text{Sub})^{12}}. \tag{1}$$

The dashes separate three *groups* on each line and are structural, not subtraction. Concretely:

4.1 Numerator — three parts

- **Planet:** its placement cusp gets power 2; each cusp it owns gets power 1.
- **Star lord:** the star lord's placement cusp gets power 4; each cusp it owns gets power 3.
- **Sub lord:** the sub lord's placement cusp gets power 6; each cusp it owns gets power 5.

4.2 Denominator — lordship of the twelve cusps

For the Planet, its Star lord, and its Sub lord in turn, scan all twelve cusps:

- cusps of which the **Planet** is star lord get power 7; sub lord, power 10;
- cusps of which the **Star lord** is star lord get power 8; sub lord, power 11;
- cusps of which the **Sub lord** is star lord get power 9; sub lord, power 12.

4.3 The twelve powers as one grid

The twelve powers are not arbitrary. They form a $3 \times 2 \times 2$ grid: the role (Planet, Star, Sub), the aspect (numerator = the point's own placement/ownership; denominator = its lordship over cusps), and the sub-aspect (placement/star-lordship vs. ownership/sub-lordship).

Table 2: The complete power assignment. Reading the two columns of each block in order Planet, Star, Sub gives $2, 1$; $4, 3$; $6, 5$ (numerator) and $7, 10$; $8, 11$; $9, 12$ (denominator).

Role	Numerator (placement / own)	Denominator (star lord / sub lord of cusps)
Planet	2 / 1	7 10
Star lord	4 / 3	8 11
Sub lord	6 / 5	9 12

Corrected source typo. The `Guide_Step6` sheet, cell B44, writes the star-lord ownership rule as "current cuspal position … power 4", duplicating part 2(a). The worked example on `Sheet2` makes the intended rule unambiguous: star-lord *ownership* carries power 3. The implementation follows the worked example, and this choice is confirmed by the 27/27 agreement with the hand-calculated charts (Section 7).

4.4 Node exception

When a node (Rahu or Ketu) appears as the Planet, Star, or Sub of a term, the *position* power is assigned to both the node's placement *and* its dispositor's placement, and the *ownership* power to the dispositor's owned cusps. This is the same dispositor substitution used for the cusp-number vectors, applied inside each numerator part.

4.5 Merging

Finally, identical base numbers combine by *adding* their powers — first within the numerator, then within the denominator, then across the two. The result is a single line of $\texttt{base}^{\text{power}}$ terms in ascending base order. The application shows this as four steps: substitute the cusp numbers, collect the numerator, collect the denominator, and merge-and-sort across both.

5 Worked Example

Figure 1 shows the reference chart ("Test1") laid out in East Indian style, with each cusp and planet placed in its sign box and annotated with its star/sub lord and, for planets, the occupied house.

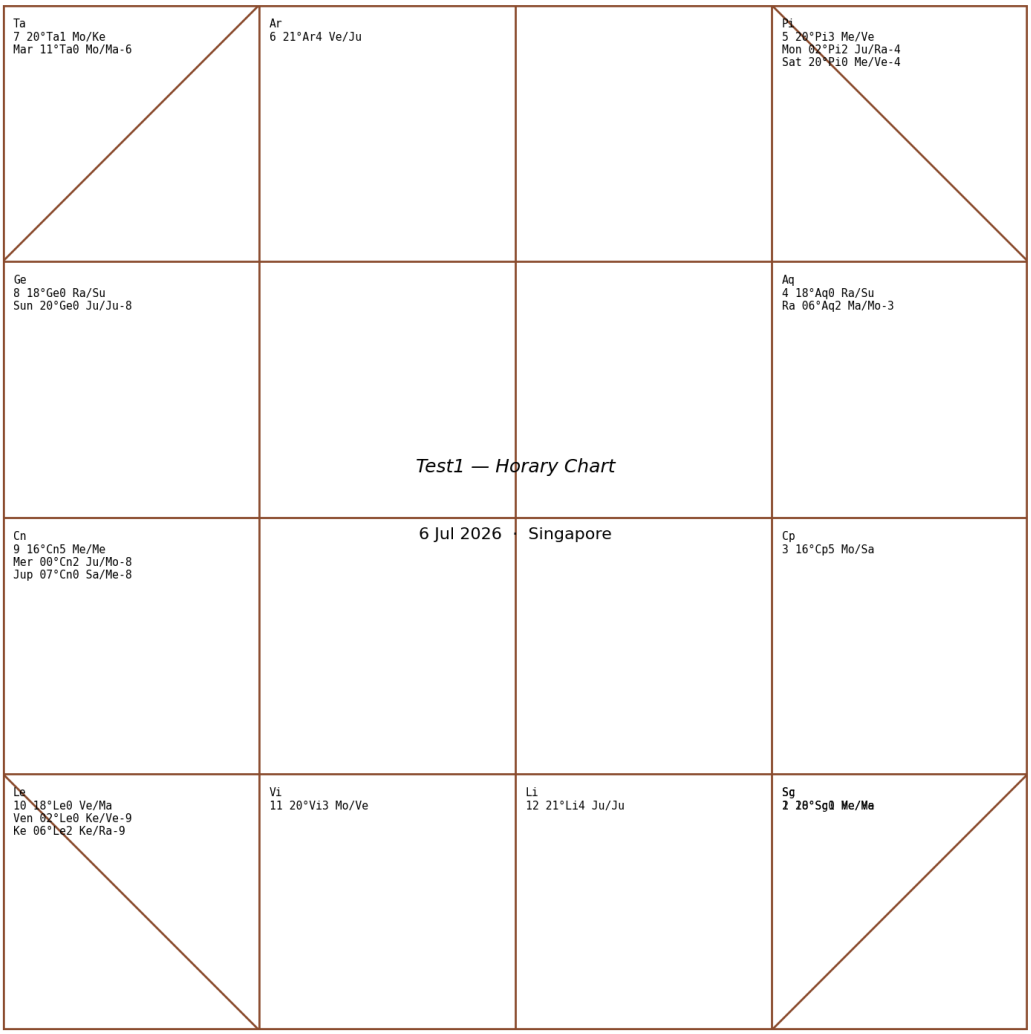


Figure 1: The reference chart rendered by Kala Chakra. Sign positions are fixed; cusp numbers and planets move with the moment of the query. Each entry reads *number/name, longitude, star lord / sub lord*, with a trailing *-house* tag for planets.

Taking the Moon as an example: it lies at 02°Pi28′13″, whose star lord is Jupiter and sub lord is Rahu, in house 4. Its cusp-number vector is $4, 9$ (placed in cusp 4, owns cusp 9). Running Equation (1) and merging yields the final power line

$$2^{3}\ 3^{18}\ 4^{22}\ 5^{3}\ 6^{11}\ 7^{7}\ 8^{13}\ 9^{1}\ 11^{7}\ 12^{19},$$

which matches the value obtained by hand. The application shows the full substitution and both collection steps that lead here.

6 Software Architecture

6.1 Two engines, one specification

The method is implemented twice: once in Python (`kp_core.py` and helpers) and once in JavaScript (`web/kp_logic.js`). The Python engine drives the command-line tools and the Excel pipeline; the JavaScript engine drives the browser app, which needs no installation. Both are held to the same specification by a parity test that runs every chart through both and compares every star lord, sub lord, house tag, cusp-number vector and power string.

6.2 Style-safe Excel editing

The East Indian chart in the output workbook is a real drawing — twelve line shapes. General-purpose spreadsheet libraries silently discard such drawings on save. Kala Chakra therefore edits workbooks *surgically*: it rewrites only the changed cells (as inline strings) directly in the underlying XML, leaving every other byte — drawings, formatting, images — untouched. The chart template is the author's own `Guide_Step2` sheet, extracted once at the zip level to preserve its twelve chart lines exactly.

6.3 Deliverables

Table 3: The main components and how they are used.

Component	Purpose
`src/kala_chakra/kp_core.py`	Engine: longitudes, star/sub lords, placement, ownership, powers
`src/kala_chakra/sf_parser.py`	Cleans a raw Solar Fire report into a chart (steps 1–3)
`excel_builder.py` (CLI `kp-build`)	Raw export → finished workbook (all steps)
`process.py` (`kp-annotate`)	Annotates an existing chart workbook in place
`validate_batch.py` (`kp-validate`)	Audits a folder of manual charts against the engine
`web/dist/Kala Chakra.html`	The standalone app (form → chart, tables, powers, Excel)
`web/dist/KP Annotator.html`	Zero-install drag-and-drop annotator

7 Validation

The engine is checked against the author's existing hand-calculated charts, which serve as ground truth. Across the full corpus:

- **Star/sub lords:** over 3,300 cusp and planet annotations reproduced exactly.
- **House placements:** over 640 diagram house-tags reproduced exactly.
- **Cusp-number vectors:** all 27 stored vectors reproduced exactly.
- **Power strings:** all 27 stored power lines reproduced exactly, including the node cases.
- **Cross-engine parity:** Python and JavaScript agree on every comparison across all 80 charts.

The only disagreements ever surfaced were three arithmetic slips in the manual charts themselves — a table row mis-read and two near-boundary house mis-assignments — each independently confirmed by the author. This is exactly the class of error the automation removes.

A regression suite (`run_tests.py`) re-runs the engine, the round-trip Excel writer, the raw-export reconstruction and the generated-table spot checks after any change; `validate_batch.py` re-audits the whole chart folder, now including power strings. Both should be run before trusting a modified build.

8 Usage

8.1 The app (no installation)

Open `web/dist/Kala Chakra.html` in any browser. Enter the question, name, date, time and location; paste the raw Solar Fire report (or drop the exported `.xlsx`); press **Calculate**. The app renders the chart, the cusp and planet tables, the step-by-step power derivation, and offers the finished Excel workbook and a print/PDF export. Everything runs locally; nothing is uploaded.

8.2 The command line (Python 3.10+)

```
pip install .                        # installs kp-build, kp-annotate, kp-
    validate

kp-build raw_export.xlsx --question "Will␣I␣...?" --name Amriit \
         --date "6␣Jul␣2026" --time "9:03␣pm" --location Singapore

kp-annotate "My␣Chart.xlsx"          # annotate an existing chart workbook
kp-validate "path/to/folder"         # audit a folder of manual charts
```

9 Limitations and Future Work

- **Ephemeris.** Longitudes are taken from external software. A future phase could compute them directly (Swiss Ephemeris with the KP ayanamsa), removing the software dependency and enabling full input from date/time/place alone.
- **Two cusps in one sign.** Placement and the span rule are handled automatically, but such charts are flagged for a human glance because the interpretive convention can vary. The encoded convention (lord owns the house the sign lies within) matched every chart tested.
- **Report format.** The raw parser targets the Solar Fire Chart Analysis Report. A different program's export would need its own small parser, though the downstream engine is unchanged.
- **Interpretation.** The system stops at the power line. Dasha timing and the final judgement remain with the astrologer, by design.

Acknowledgement of Method

The method documented here is the traditional KP (Krishnamurti Paddhati) horary technique as practised and taught within the author's family. The star/sub lord tables, the East Indian layout, and the power calculation were transcribed from the author's working notes and his father's handwritten tables. This software encodes that method faithfully; it originates no astrological doctrine.

AI tool declaration. The software and this documentation were developed with assistance from an AI system (Claude). All astrological rules originate from the author's manual workflow and were validated against his existing hand-calculated charts; the AI's role was implementation, decoding of the rule set, cross-validation, and drafting.